

Title

Impact of laughter-based physical activity on stress perception and exercise capacity in chemotherapy-treated lung cancer patients

Abstract

Background: Psychosocial stress and functional decline commonly affect lung cancer patients during chemotherapy. **Methods:** 84 lung cancer patients undergoing active chemotherapy were randomly assigned to a laughter-based exercise intervention (n=42) or routine care (n=40). The intervention involved structured sessions combining movement and laughter practices. Perceived stress, positive psychological resources, and exercise capacity were measured at baseline and post-intervention. **Results:** 80 patients (40 per group) completed the study. Patients in the laughter-based intervention demonstrated significantly higher positive psychological resources scores compared to routine care (Cohen's $d=0.71$; $p<0.05$), increased exercise tolerance ($d=0.64$; $p<0.05$), and notably lower perceived stress ratings ($d=1.09$; $p<0.05$). Within-group improvements in stress occurred in both arms, but between-group differences favored the intervention. **Conclusions:** Laughter-based aerobic activities appear effective in reducing stress perception and enhancing exercise tolerance in chemotherapy patients, suggesting potential adjunctive value for improving psychological well-being during active cancer treatment.